

AWS Global Accelerator – Multi-Region Web Routing

To Begin with the Lab

Summary of the Lab

In this lab, a complete multi-region web application architecture was built using **Amazon VPC**, **Amazon EC2**, **Application Load Balancer**, and **AWS Global Accelerator**. Separate environments were deployed in the Mumbai and N. Virginia regions with dedicated networking components including public and private subnets, Internet Gateways, NAT Gateways, and route tables. Web servers were launched inside private subnets and registered behind regional Application Load Balancers. Finally, **AWS Global Accelerator** was configured to route users to the nearest healthy regional endpoint and automatically redirect traffic during regional failures. The setup was verified by testing regional routing, health checks, and failover behavior.

Understanding AWS Global Accelerator and Multi-Region Web Routing

AWS Global Accelerator is a networking service that provides users with a single global entry point for applications deployed across multiple AWS Regions. Instead of users connecting directly to regional load balancers, they connect to Global Accelerator, which routes traffic through the AWS global network to the nearest healthy endpoint. This improves latency, increases availability, and provides automatic failover when a regional endpoint becomes unavailable. In simple terms, it helps a global website feel faster and stay online even when traffic must move between regions.

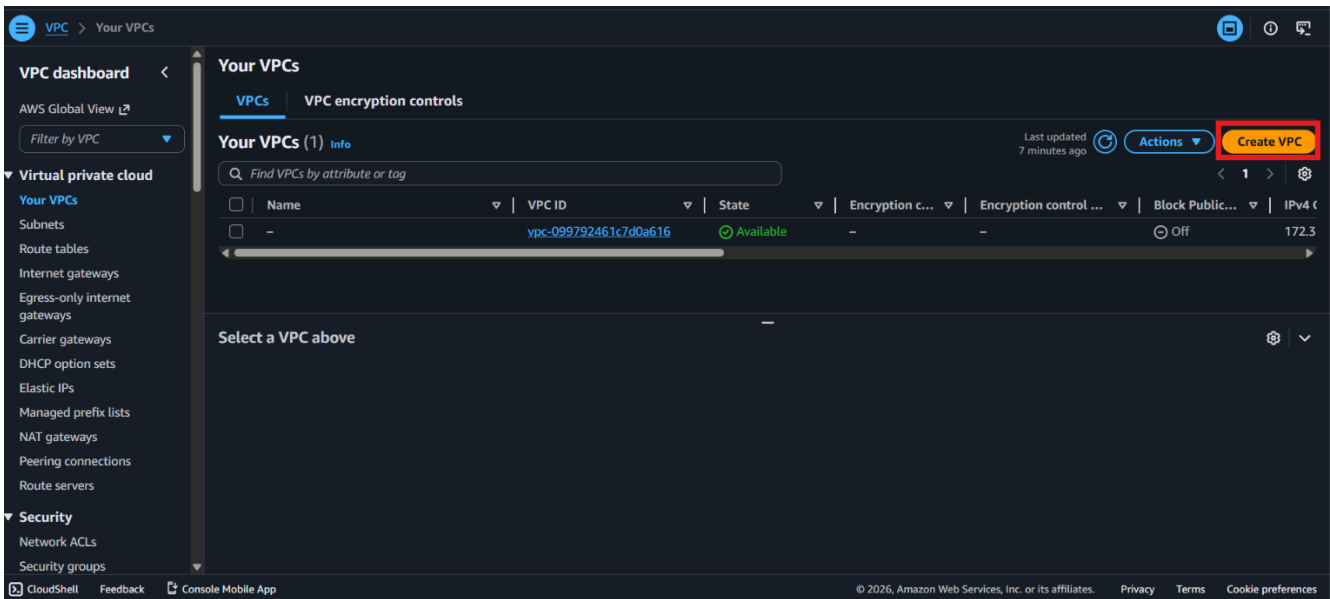
Example:

An e-commerce company hosts its application in Mumbai and Virginia using two Application Load Balancers. Customers from India are automatically routed to the Mumbai endpoint while customers from North America are routed to the Virginia endpoint. If the Mumbai region becomes unavailable, **AWS Global Accelerator** immediately redirects Indian users to Virginia without changing the application URL.

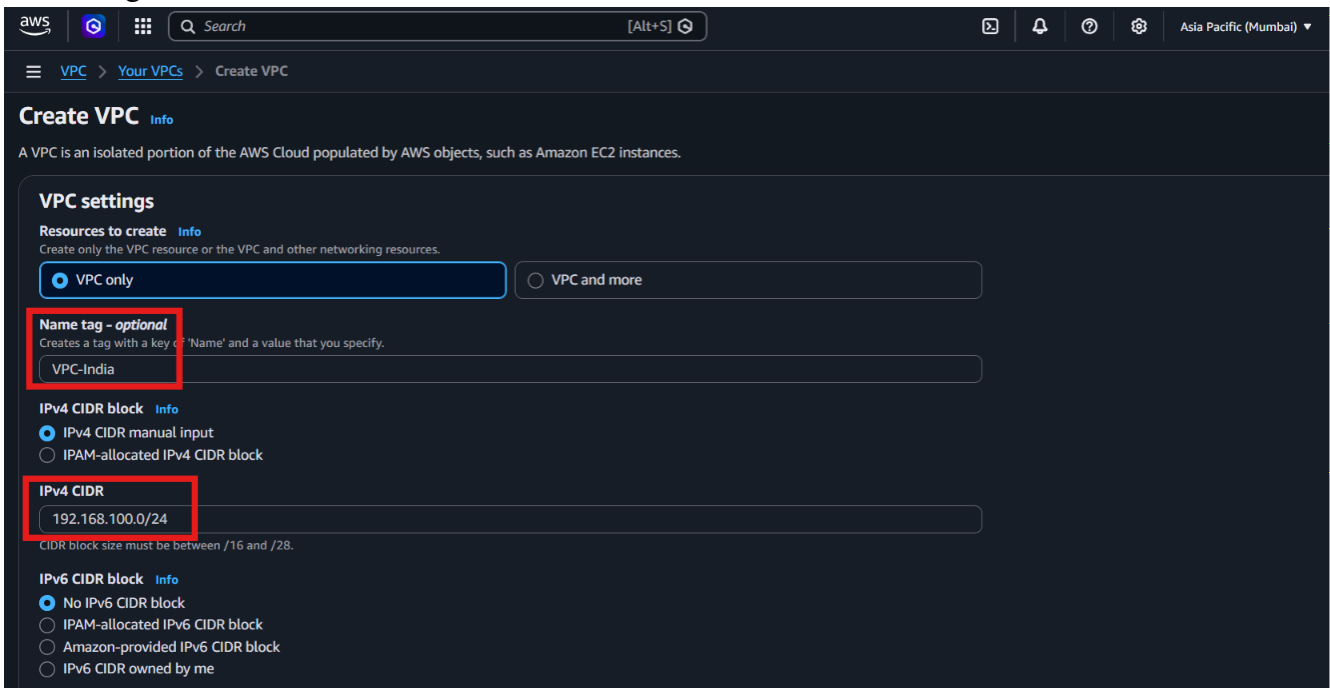
Lab Steps

1. Configuration of VPC's in Respective Regions

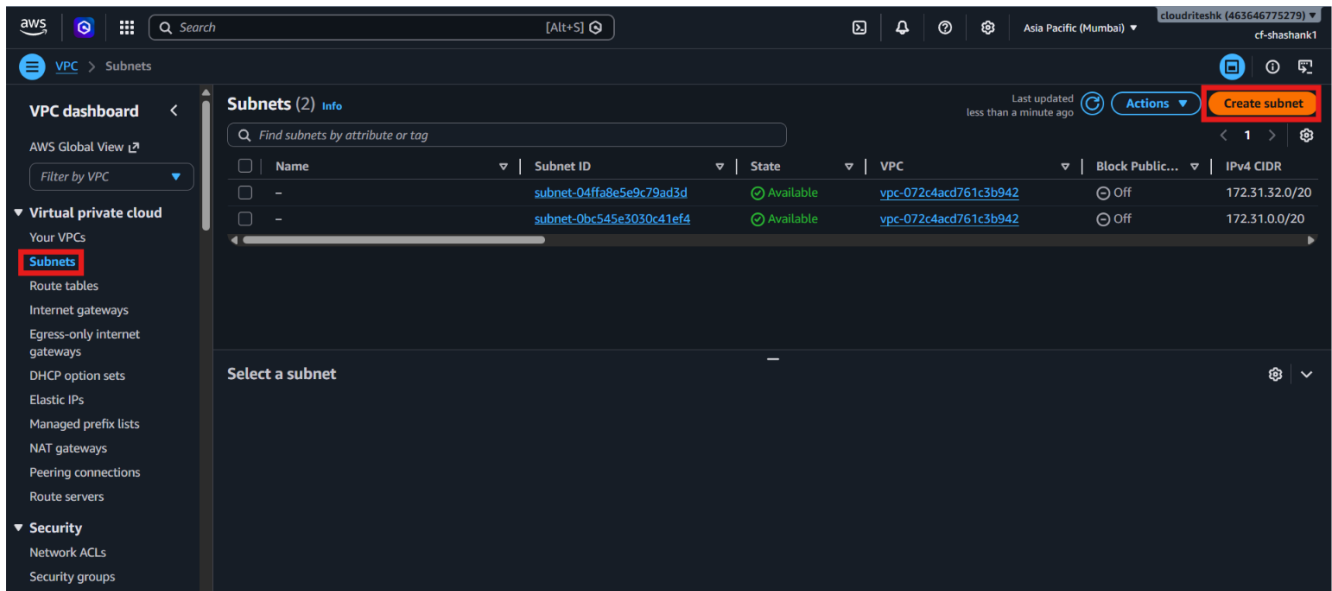
- Sign in to the **AWS Management Console**.
- Switch to the **Mumbai Region (ap-south-1)**.
- Navigate to **Amazon VPC → Your VPCs**.
- Delete the default VPC to avoid confusion during the lab.
- Click **Create VPC**.



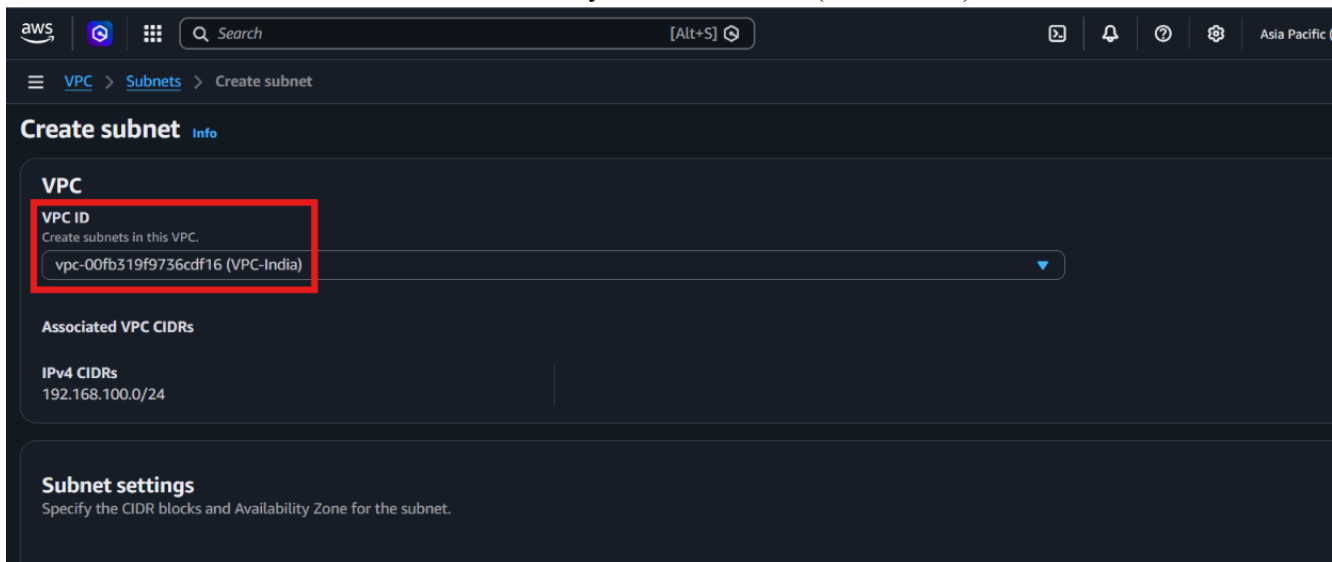
- Create a VPC named **VPC-India**.
- Configure the CIDR block as 192.168.100.0/24 and click **Create VPC**.



- Navigate to **Amazon VPC** → **Subnets**.



- Click **Create Subnet** and select **VPC ID** you have created (VPC-India).



- Create **Public-Subnet-1A** in **ap-south-1a** with CIDR 192.168.100.0/26.

aws [Search] [Alt+S]

VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 4

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Public-Subnet-1A
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1 (ap-south-1a)

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.100.0/24

IPv4 subnet CIDR block
192.168.100.0/26 64 IPs

Tags - optional

Key	Value - optional
-----	------------------

- Create **Public-Subnet-1B** in **ap-south-1b** with CIDR 192.168.100.64/26.
- Create **Private-Subnet-1A** in **ap-south-1a** with CIDR 192.168.100.128/26.
- Create **Private-Subnet-1B** in **ap-south-1b** with CIDR 192.168.100.192/26.

aws [Search] [Alt+S]

VPC > Subnets > Create subnet

Subnet settings

Subnet 4 of 4

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Private-Subnet-1B
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Mumbai) / ap-south-1 (ap-south-1b)

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.100.0/24

IPv4 subnet CIDR block
192.168.100.192/26 64 IPs

Tags - optional

Key	Value - optional
Name	Private-Subnet-1B

[Add new tag](#)

- Click **Create subnet**.
- Navigate to **Amazon VPC** → **Route Tables**.
- Rename the default route table to **RT-Private**.

The screenshot shows the AWS Management Console interface for the 'Route tables' section. The left sidebar contains the 'VPC dashboard' and 'Virtual private cloud' sections. The main content area displays a list of route tables. The first route table, 'RT For Private', is selected and highlighted with a red box. Below the list, the details for this route table are shown, including its ID, VPC, and other attributes.

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
RT For Private	rtb-04d095bca083d95ac	-	-	Yes	vpc-00fb319f9736cdf16 VPC-India

Details for 'rtb-04d095bca083d95ac / RT For Private':

- Route table ID: rtb-04d095bca083d95ac
- Main: Yes
- Explicit subnet associations: -
- Edge associations: -
- VPC: vpc-00fb319f9736cdf16 | VPC-India
- Owner ID: 463646775279

- Create a new route table named **RT-Public**.

The screenshot shows the 'Create route table' page in the AWS Management Console. The 'Route table settings' section is visible, with the 'Name' field set to 'RT For Public' and the 'VPC' dropdown set to 'vpc-00fb319f9736cdf16 (VPC-India)'. The 'Tags' section is also visible, showing a tag with the key 'Name' and the value 'RT For Public'. The 'Create route table' button is highlighted with a red box.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
RT For Public

VPC
The VPC to use for this route table.
vpc-00fb319f9736cdf16 (VPC-India)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key: Name Value - optional: RT For Public

Buttons: Add new tag, Cancel, Create route table

- Click **RT For Public** → **Subnet associations** → **Edit subnet associations**.

Route tables (1/3) Info

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-017299725a38d5666	2 subnets	-	Yes	vpc-072c4acd761c3b942
RT For Private	rtb-04d095bca083d95ac	-	-	Yes	vpc-00fb319f9736cdf16 VPC-I.
RT For Public	rtb-07ebaa72d98f50422	-	-	No	vpc-00fb319f9736cdf16 VPC-I.

rtb-07ebaa72d98f50422 / RT For Public

Details Routes **Subnet associations** Edge associations Route propagation Tags

Explicit subnet associations (0)

[Edit subnet associations](#)

No subnet associations
You do not have any subnet associations.

- Associate the public subnets with **RT-Public**.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒ Public-Subnet-1A	subnet-033e2e19c569f27fa	192.168.100.0/26	-	Main (rtb-04d095bca083d95ac / RT Fo...
☐ Private-Subnet-1B	subnet-051e4e384838f4b9	192.168.100.192/26	-	Main (rtb-04d095bca083d95ac / RT Fo...
☒ Public-Subnet-1B	subnet-05681b26191591078	192.168.100.64/26	-	Main (rtb-04d095bca083d95ac / RT Fo...
☐ Private-Subnet-1A	subnet-012b986e32a7dafbe	192.168.100.128/26	-	Main (rtb-04d095bca083d95ac / RT Fo...

Selected subnets

subnet-033e2e19c569f27fa / Public-Subnet-1A × subnet-05681b26191591078 / Public-Subnet-1B ×

[Cancel](#) [Save associations](#)

- Keep the private subnets associated with **RT-Private**.
- Navigate to **Internet Gateways** and click **Create internet gateway**.

Internet gateways (2) Info

Name	Internet gateway ID	State	VPC ID	Owner
netwest-igw1	igw-075c414370aa73341	Detached	-	463646775279
asd1	igw-0e8284fa57b179741	Attached	vpc-072c4acd761c3b942	463646775279

Select an internet gateway above

[Create internet gateway](#)

- Create an Internet Gateway named **IG**.

aws [Search] [Alt+S] Asia Pacific (Mumbai) cloudrishik (46)

VPC > Internet gateways > Create internet gateway

Create internet gateway Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

IG

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key Value - optional

Q Name X Q IG X Remove

Add new tag

You can add 49 more tags.

Cancel Create internet gateway

- Click **Attach VPC** and attach the Internet Gateway to **VPC-India**.

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VPC > Internet gateways > Attach to VPC (igw-05c9d506143f906ec)

The following internet gateway was created: igw-05c9d506143f906ec - IG. You can now attach to a VPC to enable the VPC to communicate with the internet. Attach to a V

Attach to VPC (igw-05c9d506143f906ec) Info

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the Internet gateway to this VPC.

Q vpc-00fb319f9736cdf16 X

Use: "vpc-00fb319f9736cdf16"

vpc-00fb319f9736cdf16 - VPC-India

Cancel Attach internet gateway

- Open **RT-Public** and add a route for 0.0.0.0/0 pointing to the **NAT Gateway**.

aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) cf-shashank1

VPC > Route tables > rtb-07ebaa72d98f50422 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
192.168.100.0/24	local	Active	No	CreateRouteTable
Q 0.0.0.0/0 X	Q NAT Gateway X	Active	No	CreateRoute

Add route

Use: "nat-"

nat-0533d87c7b436ff91 (Nat Gateway India)

Cancel Preview Save changes

- Navigate to **NAT Gateways** and click on **Create NAT Gateway**.
- Choose Availability Zone as **Zonal**.

- Choose **Public-Subnet-1A** in **Subnet**.
- Allocate an Elastic IP for the NAT Gateway.

aws [Search] [Alt+S] Asia Pacific (Mumbai) cloud

VPC > NAT gateways > Create NAT gateway

🟢 Elastic IP address 65.2.196.236 (eipalloc-00fc2485648911b57) allocated.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
Nat Gateway India
The name can be up to 256 characters long.

Availability mode [Info](#)
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ Regional - new
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☒ Zonal
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.
subnet-033e2e19c569f27fa (Public-Subnet-1A)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.
eipalloc-00fc2485648911b57 [Allocate Elastic IP](#)

▶ **Additional settings** [Info](#)

- Wait until the NAT Gateway becomes available.
- Open **RT-Private** and add a default route for 0.0.0.0/0 pointing to **Nat Gateway India**.
- Repeat the same VPC, subnet, route table, Internet Gateway, and NAT Gateway setup in the **N. Virginia Region (us-east-1)**.
- Use **VPC-USA** with CIDR 192.168.200.0/24 for the US region.
- Verify that both regions have one VPC, two public subnets, two private subnets, a public route table, a private route table, an Internet Gateway, and a NAT Gateway.

2. Configuration of Security Groups

- Click on **EC2** → **Security Groups** → **Create security group**.
- In both regions, create an **ALB-SG** security group that allows HTTP traffic from anywhere on port 80, so configure the name, description and **VPC**.

aws [Search] [Alt+S] Asia Pacific (Mumbai) cloud

EC2 > Security Groups > Create security group

Create security group

[Info](#)
A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)
ALB-SG
Name cannot be edited after creation.

Description [Info](#)
Secure ALB

VPC [Info](#)
vpc-00fb319f9736cdf16 (VPC-India)

- Choose proper **Inbound** and **Outbound** rule as shown and click on **Create security group**.

The screenshot shows the 'Create security group' page in the AWS Management Console. The 'Inbound rules' section is active, showing a rule for HTTP (Type: HTTP, Protocol: TCP, Port range: 80, Source: Anywh..., 0.0.0.0/0). The 'Outbound rules' section is also visible, showing a rule for All traffic (Type: All traffic, Protocol: All, Port range: All, Destination: Custom, 0.0.0.0/0). Both rules are highlighted with red boxes.

- In both regions, create a **Web-Server-SG** security group that allows HTTP traffic only from the ALB security group with **Inbound and Outbound Rules** as shown.
- Select **ALB-SG** as the **Source** in **Inbound Rule** while creating Web-Server-SG.
- Add an SSH inbound rule to **WebServer-SG** from 0.0.0.0/0 so the instances can be managed only if needed.

The screenshot shows the 'Edit inbound rules' page for the 'Web-Server-SG' security group. The 'Inbound rules' section is active, showing two rules: one for HTTP (Type: HTTP, Protocol: TCP, Port range: 80, Source: Cus..., sg-086d15fec408) and one for SSH (Type: SSH, Protocol: TCP, Port range: 22, Source: Any..., 0.0.0.0/0). Both rules are highlighted with red boxes.

- Launch **Web-Server-1** in the India private subnet (Private-Subnet-1A) using **Amazon Linux, t3.micro, VPC-India, Private-Subnet-1A, and WebServer-SG** as shown below.
- Make sure to disable **Auto-assign public IP**.

Name and tags Info

Name

Web-Server-1

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

My AMIs

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

Canonical Ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-0db56f446d44f2f09 (64-bit (x86), uefi-preferred) / ami-07c18f602ca3306b9 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12.....[read more](#)

ami-0db56f446d44f2f09

Virtual server type (instance type)

t3.micro

Firewall (security group)

Web-Server-SG

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro

Cancel

Launch instance

Preview code

aws

Search

[Alt+S]

EC2

Instances

Launch an instance

EC2

Dashboard

AWS Global View

Events

Instances

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Network settings Info

VPC - required Info

vpc-0d1bb5239b4840bd3 (VPC-India)

192.168.100.0/24

Subnet Info

subnet-068e5cfb395f62ccf

Private-Subnet-1A

VPC: vpc-0d1bb5239b4840bd3 Owner: 463646775279

Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone

IP addresses available: 59 CIDR: 192.168.100.128/26

Auto-assign public IP Info

Disable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups Info

Select security groups

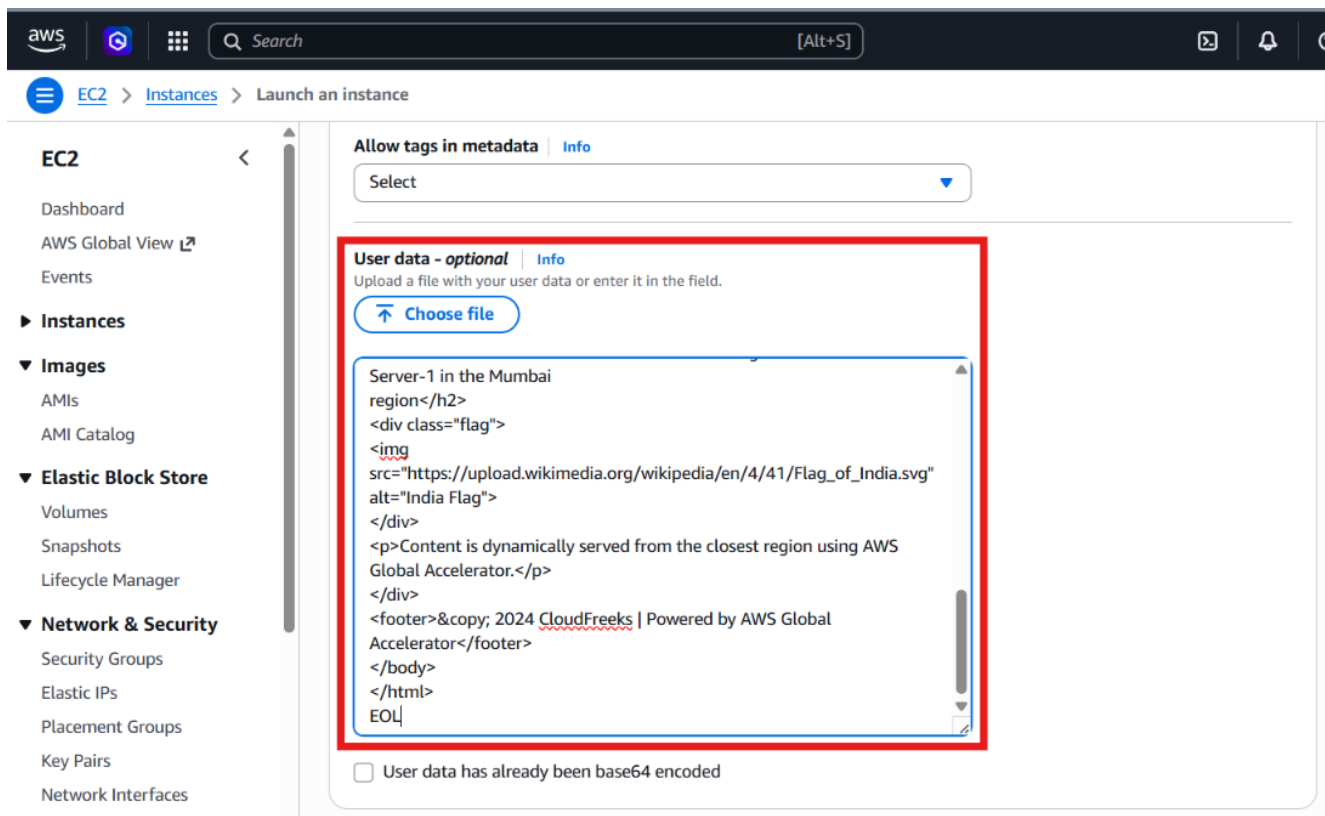
Web-Server-SG sg-0e09f21f06c9c3039

VPC: vpc-0d1bb5239b4840bd3

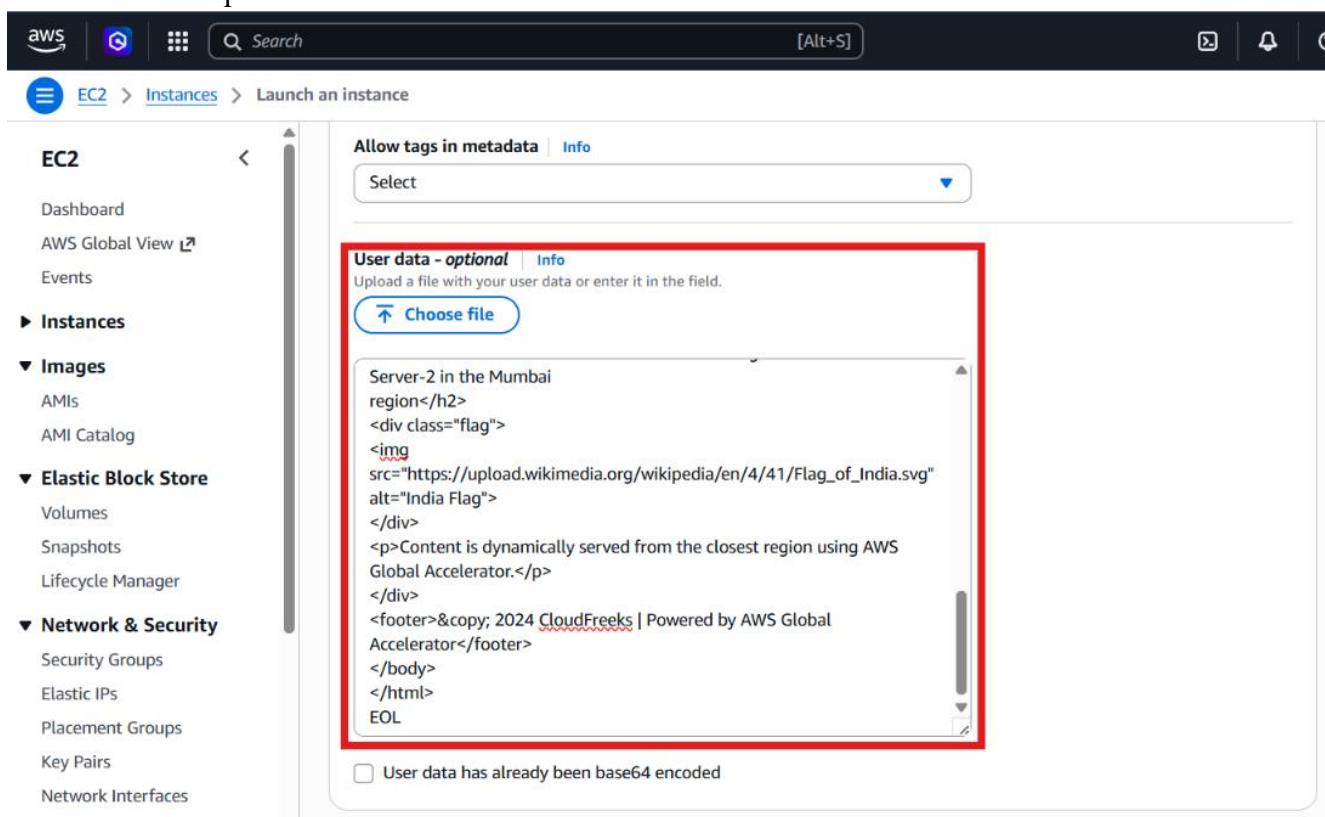
Compare security group rules

Advanced network configuration

- Paste the provided Mumbai user data script to install Apache and serve region-specific HTML.



- Launch **Web-Server-2** in the India private subnet with the same configuration and the Mumbai user data script.



- Repeat the same two EC2 launches in the US region using the US user data scripts.
- Navigate to **Target Groups** and create **TG-India** with HTTP port 80 in **VPC-India**.

aws [Search] [Alt+S] Asia Pacific (Mumbai) cloudriteshk (4636467752) cf-shash

EC2 > Target groups > Create target group

AMI Catalog

- ▼ Elastic Block Store
 - Volumes
 - Snapshots
 - Lifecycle Manager
- ▼ Network & Security
 - Security Groups
 - Elastic IPs
 - Placement Groups
 - Key Pairs
 - Network Interfaces
- ▼ Load Balancing
 - Load Balancers
 - Target Groups
 - Trust Stores
- ▼ Auto Scaling
 - Auto Scaling Groups

Settings

Target group name
Name must be unique per Region per AWS account.
TG-India
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 8/32

Protocol
Protocol for communication between the load balancer and targets.
HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
80
1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.
☒ IPv4
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
☐ IPv6
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.
vpc-0d1bb5239b4840bd3 (VPC-India)
192.168.100.0/24 [Create VPC](#)

- Register both India EC2 instances as targets and click **Include as pending below** and click **Next**.

EC2 > Target groups > Create target group

Images

- AMIs
- AMI Catalog

Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

Load Balancing

- Load Balancers
- Target Groups
- Trust Stores

Auto Scaling

- Auto Scaling Groups

Settings

Step 1: Create target group
Step 2 - recommended: **Register targets**
Step 3: Review and create

Register targets - recommended
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Filter instances

☒	Instance ID	Name	State	Security groups	Zone
☒	i-00193c80bf57b50fe	Web-Server-2	Running	Web-Server-SG	ap-so
☒	i-0b2dc2d16c581637f	Web-Server-1	Running	Web-Server-SG	ap-so

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.
80
1-65535 (separate multiple ports with comma)

[Include as pending below](#)

Review targets

- Wait for the target health to become healthy.
- Now go to **Load Balancer** and choose **Application Load Balancer**.

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EC2 > Load balancers > Compare and select load balancer type

Application Load Balancer Info

Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

Network Load Balancer Info

Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

Create

Gateway Load Balancer Info

Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

Create

- Create an Application Load Balancer named **ALB-India** in the public subnets of **VPC-India** as shown below.

aws [Alt+S] Asia Pacific (Mumbai)

EC2 > Load balancers > Create Application Load Balancer

EC2

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Lifecycle Manager

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Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

ALB-India

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme Info

Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type Info

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**

Includes only IPv4 addresses.

☐ **Dualstack**

Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **Internet-facing** load balancers only.

Network mapping Info

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC Info

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0d1bb5239b4840bd3 (VPC-India)
192.168.100.0/24

Create VPC

EC2 > Load balancers > Create Application Load Balancer

Network mapping Info
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0d1bb5239b4840bd3 (VPC-India)
192.168.100.0/24

IP pools Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancer IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ ap-south-1a (aps1-az1)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 5 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.
subnet-05dc525c5d56d0446 (Public-Subnet-1A)
192.168.100.0/26
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (/)

☒ ap-south-1b (aps1-az3)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 5 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.
subnet-0827e46e96a25e093 (Public-Subnet-1B)
192.168.100.64/26
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (/)

- Attach ALB-SG security group to the load balancer.

EC2 > Load balancers > Create Application Load Balancer

Security groups Info
A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.

Security groups
Select up to 5 security groups

ALB-SG (sg-05f31770a2c68a9f5) ✕
Inbound rule (1) Outbound rule (1)

Listeners and routing Info
A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80 Remove

Protocol: HTTP Port: 80
1-65535

Default action Info
The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action
☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group Info
Choose a target group and specify routing weight or [create target group](#).

Target group: TG-India HTTP Weight: 1 Percent: 100%
0-999

Target group stickiness Info
Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

- Configure the listener on HTTP 80 and forward traffic to TG-India.

EC2 > Load balancers > Create Application Load Balancer

Listeners and routing Info
A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80 Remove

Protocol: HTTP Port: 80
1-65535

Default action Info
The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action
☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group Info
Choose a target group and specify routing weight or [create target group](#).

Target group: TG-India HTTP Weight: 1 Percent: 100%
0-999

Target group stickiness Info
Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional

- Create the matching US target group and Application Load Balancer in VPC-USA.
- Verify the India ALB DNS name opens the website and alternates between the two India web servers.

- Verify the US ALB DNS name opens the USA website and alternates between the two US web servers.

3. Configure AWS Global Accelerator and test failover

- Navigate to **AWS Global Accelerator** and click **Create Accelerator**.

The screenshot shows the AWS Global Accelerator console landing page. On the left is a navigation menu with 'AWS Global Accelerator', 'Accelerators', and 'Cross-account attachments'. The main content area has a dark header with the text 'Networking & Content Delivery' and 'AWS Global Accelerator Improve availability and performance for your applications'. Below this is a 'How it works' diagram showing traffic flow from users to an accelerator and then to endpoints. On the right, there are two cards: 'Create accelerator' with a 'Create accelerator' button, and 'Speed Test Tool' with a 'Speed Comparison Tool' link.

- Create a standard accelerator named **My-GA** using **IPv4**.

The screenshot shows the 'Create accelerator' page in the AWS Global Accelerator console. The breadcrumb trail is 'AWS Global Accelerator > Accelerators > Create accelerator'. The left sidebar shows 'AWS Global Accelerator', 'Accelerators', and 'Cross-account attachments'. The main content area is titled 'Step 4 Add endpoints' and 'Basic configuration'. It contains three main sections: 'Accelerator name' with a text input field containing 'My-GA', 'Accelerator type' with radio buttons for 'Standard' (selected) and 'Custom routing', and 'IP address type' with a dropdown menu set to 'IPv4'. There are also expandable sections for 'IP address pool selection' and 'Tags'. At the bottom right, there are 'Cancel' and 'Next' buttons.

- Add a listener using protocol **TCP** and port **80**.

The screenshot shows the 'Add listeners' page in the AWS Global Accelerator console. The breadcrumb trail is 'AWS Global Accelerator > Accelerators > Create accelerator'. The left sidebar shows 'AWS Global Accelerator', 'Accelerators', and 'Cross-account attachments'. The main content area is titled 'Step 1 Enter name' and 'Add listeners'. It contains a 'Listeners' section with a text input for 'Ports' containing '80', a dropdown for 'Protocol' set to 'TCP', and a dropdown for 'Client affinity' set to 'None'. There is a 'Remove' button and an 'Add listener' button. At the bottom right, there are 'Cancel', 'Previous', and 'Next' buttons.

- Create an endpoint group for **Mumbai (ap-south-1)** and **N. Virginia (us-east-1)** with traffic dial set to 100%.

AWS Global Accelerator > Accelerators > Create accelerator

Step 3: Add endpoint groups

Add endpoint groups

An accelerator includes one or more listeners that direct traffic to one or more endpoint groups. An endpoint group includes endpoints, such as load balancers.

Listener: 80 TCP

Each listener can have multiple endpoint groups. Each endpoint group can only include endpoints that are in one Region. You aren't required to add an endpoint group, but until you do, traffic to this listener won't reach any endpoints.

Endpoint group 1

Region: **ap-south-1** Traffic dial: 100 **Remove**

Configure port overrides

Configure health checks

Endpoint group 2

Region: **us-east-1** Traffic dial: 100 **Remove**

Configure port overrides

Configure health checks

Add endpoint group

Next

- Add endpoints and endpoint type as **Application Load Balancer** and its endpoint as shown for both the regions.

AWS Global Accelerator > Accelerators > Create accelerator

Step 4: Add endpoints

Add endpoints

Now that you've added one or more endpoint groups, you can add endpoints to each one. If you don't have any endpoints yet, create one on the Elastic Load Balancing (ELB) console or create an EC2 instance. Endpoints must be in the same Region as the endpoint group. You must have the required permissions to view available endpoint resources.

Listener: 80 TCP

Global Accelerator routes traffic that arrives on these ports to endpoints in regional endpoint groups. All endpoints for an endpoint group must be in the same Region.

Client IP address preservation is supported, with some limitations, for Network Load Balancers with security groups

You cannot enable client IP address preservation when you add an endpoint that is a dual-stack Network Load Balancer or a Network Load Balancer with a TLS listener attached.

Endpoint group: **ap-south-1**
Traffic dial: 100%

Endpoint 1

Add a resource specified in a cross-account attachment

Select account ID of a cross-account attachment owner

Endpoint type: **Application Load Balancer** Endpoint: **arn:aws:elasticloadbalancing...** Weight: 128 **Remove**

Preserve client IP address

Global Accelerator preserves the client IP address for internet-facing Application Load Balancers unless you clear the check box to disable the feature. All internal Application Load Balancers automatically preserve the client IP address. Make sure that your endpoints are configured to accept traffic from the preserved client IP addresses.

☒ **Preserve client IP address**

- Create the accelerator and wait for it to become enabled and deployed.
- Open the accelerator DNS name and verify that users are routed to the nearest regional load balancer.

AWS Global Accelerator > Accelerators > My-GA

My-GA ⌵ ⌵ Delete

My-GA configuration Edit

Name: My-GA

Provisioning status: Deployed

Enabled: Enabled

Static IP address set: Static IP address set

IP address: 75.2.49.185, 166.117.238.188

Address pool: Amazon, Amazon

DNS name: a72cfd047f7aed96fawsglobalaccelerator.com

Edited: Wednesday, June 10, 2026 8:50 AM GMT

ARN: arn:aws:globalaccelerator::463646775279:accelerator/45241cc5-2780-4cf7-a01e-c2c2bc2d0589

Routing type: Standard

Created: Wednesday, June 10, 2026 8:49 AM GMT

Listeners (1) View details Edit Remove Add listener

Find listeners

Listener ID	Ports and protocol	Endpoint groups	Status
arn:ad19dcba	Listener: 80 TCP	ap-south-1, us-east-1	All healthy

- Use a geographic testing tool or website like [geopeeker.com](https://www.geopeeker.com) if needed to confirm different user locations receive the correct regional content.

GeoPeeker

Viewing results for a72cfd047f7aed96fawsglobalaccelerator.com

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Location	IP Address	Latency
Singapore	166.117.238.188	1 ms
Brazil	166.117.238.188	0 ms
Virginia	166.117.238.188	1 ms

- Check endpoint health inside the accelerator and confirm both regional ALBs are healthy.
- Simulate a failure by deleting the India load balancer.

Load balancers (1/1) [What's new?](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

Name	State	Type	Scheme	IP address type
ALB-India	Active	application	Internet-facing	IPv4

Load balancer: ALB-India

[Details](#) [Listeners and rules](#) [Network mapping](#) [Resource map](#) [Security](#) [Monitoring](#) [Integrations](#) [Attributes](#) [Capacity](#)

Details

Load balancer type Application	Status Active	VPC vpc-0d1bb5239b4840bd3	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone ZP97RAFLXTNZK	Availability Zones subnet-0827e46e96a25e093 ap-south-1b (ans1-az3)	Date created June 10, 2026, 13:49 (UTC+05:30)

- Wait for the Mumbai endpoint health to become unknown or unhealthy.

Endpoint group: ap-south-1 [Remove](#)

ap-south-1 configuration [Edit](#)

Endpoint group
ap-south-1

ARN
[arn:aws:globalaccelerator:463646775279:accelerator/45241cc5-2780-4cf7-a01e-c2c2bc2d0589/listener/ad19dcba/endpoint-group/cc1c66bbf00a](#)

Threshold count
3

Traffic dial
100%

Health check interval
30 seconds

Health check path
undefined

Health check port
80

Health check protocol
TCP

Endpoints (1) [Edit](#) [Remove](#) [Add endpoint](#)

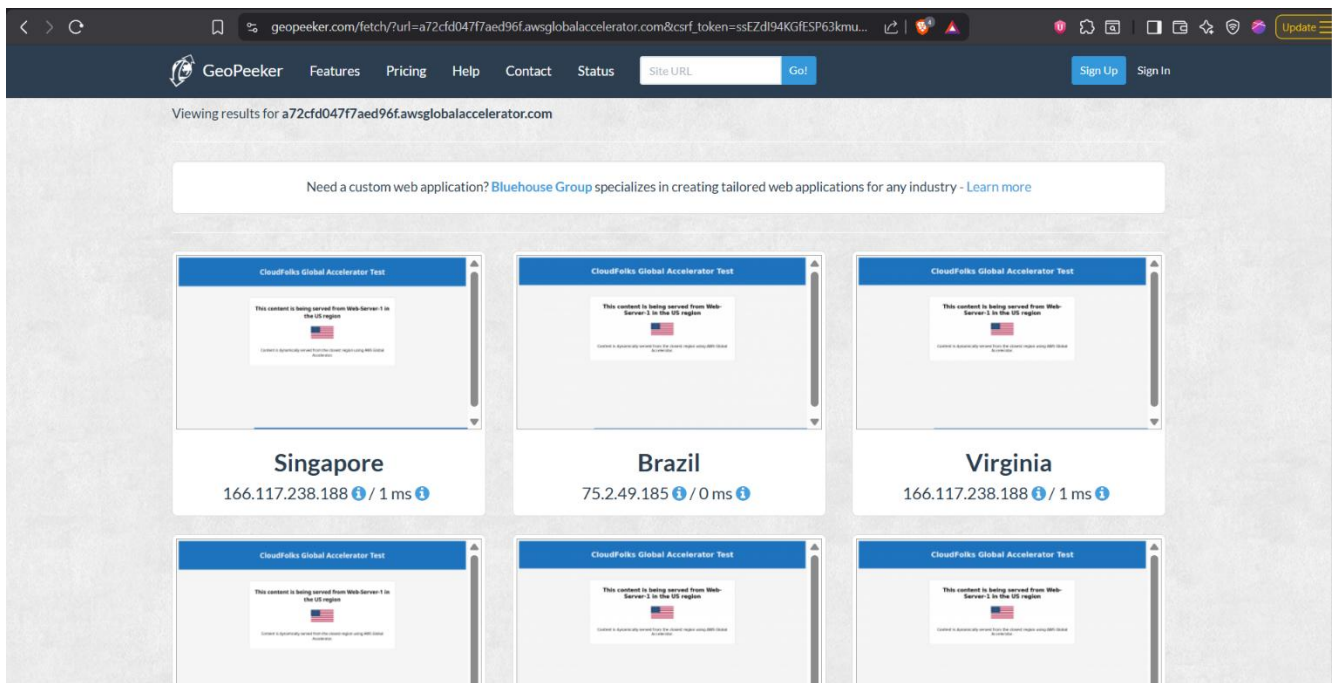
Find endpoints

Endpoint ID	Endpoint type	Client IP address	Health status	Weight
arn:aws:elasticloadbalancing:ap-south-1:463646775279:loadbalancer/app/ALB-India/a20133f940dc388b	ALB	Enabled	Unknown	128

Port overrides [Edit](#)

Find port overrides

- Open the accelerator URL again and verify traffic fails over to the Virginia region automatically.



- Recreate the deleted regional load balancer if you want to restore the original multi-region balance.
- Verify that the single Global Accelerator DNS endpoint continues to keep the application available during regional failure.